Precision-fermented dairy, egg and serum proteins: what they cost, and where evidence would help

Memo · v4.1, seven proteins (2026-11-04 13:32) · Pablo Antonio Moreno Casares · techno-economic model `pf-tea`, Humbird (2021) method, checked against NREL, Intelligen and Voutilainen designs · prepared with an AI assistant; not yet reviewed by human experts

How the price works. Figures are minimum selling prices: costs per kg of protein including a capital charge that repays the plant over 10 years at 7.5% (14.6% of the investment a year), not market prices. Base β-lactoglobulin, $21.2/kg: sugar $4.1, the plant $11.6 (that charge plus 9%/yr maintenance and insurance, over what each m³ of fermenter makes), the rest $5.5. Each m³ makes more when cells secrete faster (mg of protein per g of cells per hour); the *titer* is grams per litre at harvest.

Bottom line.

- **Today's public strains cost far too much.** At the best peer-reviewed titers, β-lactoglobulin (BLG, whey), ovalbumin (OVA), ovomucoid (both egg white) and β-casein cost $43.0, $227, $479 and $808 per kg of protein. The proteins they replace sold for $9–$18 (whey), $10–$21 (dried egg white) and $8–$19 (casein) per kg of protein in 2015–2025 (2025 dollars). Only one public titer comes near the design point, and a company reports it: EVERY's 30 g/L ovomucoid patent example, at $23.7.
- **At the titers in companies' own cost models, the commodity proteins near their markets.** At 42 g/L, BLG costs $15.8 ($12.3 with design-report capital, NREL-style), OVA $15.9, ovomucoid $16.0 and β-casein $16.4. Whey sold above the BLG figure in 3 of 21 annual export prices in 2015–2025. Casein has the thinnest evidence: no secreted casein titer is public.
- **Secretion, not titer, is the gate.** To reach $20/kg, cells must secrete about 5–7 mg of protein per g of cells per hour at the base cell density. The base design secretes 4.7. With today's biology and capital, the best design reaches $7.1/kg BLG at 40 mg/g/h and $12.0 at 10. No published **K. phaffii** process comes near either rate; the best glucose fed-batch we found reached 2.1.
- **Sugar is not the problem.** The base case's glucose ($4.1/kg BLG) costs no more than an animal's feed per kg of protein, with all the feed charged to protein ($4.7–$6.6 in the US). The gap is everything but the feedstock: $17/kg in the plant, against $2–$8 on the farm.
- **A best case, not a floor.** Set every assumed limit at its best, with 400 m³ vessels and 100 kt/yr. If cells secrete 40 mg/g/h, BLG, OVA, ovomucoid and β-casein then cost $4.1, $4.2, $4.3 and $4.5 per kg. At 10 mg/g/h BLG costs $8.1; at the best rate shown so far (2.1 mg/g/h), $24.5.
- **Scarce proteins are the easier case.** Intelligen's lactoferrin design reaches 10 g/L with methanol. At that titer, a 250 t/yr plant makes lactoferrin for $130–$168/kg ($177 costed on methanol). That is below the $250–$1,585/kg reported for dairy lactoferrin. At the best public methanol-free titer (2.8 g/L) it costs $413, inside that range.
- **The route can change the product.** Precipitated, as dairy casein is, β-casein reaches dairy purity at $22.0/kg. The only public yeast casein process uses membranes alone. Its powder is about 6% β-casein and mostly yeast carbohydrate.
- **For cultivated-meat albumin, the columns set the cost.** We use the industrial process's minimal train (cation exchange, then anion exchange), cell-culture purity and a 1 kt/yr plant. Albumin then costs $215/kg ($205 at 42 g/L; $425 at the best public titer, 8.7 g/L of broth made with methanol). Bulk bovine albumin sells for $40–$70, and GFI targets $10/kg. Capture alone would cost $143, and a $100/L resin $114. Membranes alone cost $34.3 but reach only 77% albumin. At the base cost, albumin adds $1.95–$10.23 per kg of cultivated meat, and leading firms are removing it from their media.
- **For α-lactalbumin and κ-casein, the strains are missing.** α-Lac, the infant-formula whey protein, costs $33.8/kg at the base titer on a 1 kt/yr plant. Reports put α-lac at $25–$129 per kg (market reports). At its best public titer (0.6 g/L of supernatant, human α-lac) it costs $1,848. Rennet cheese needs κ-casein at about 19% of its casein. At the best κ titer (0.25 g/L), an α_{s1} + κ blend costs $567–$573/kg. To add under 10%, κ co-precipitated with α_{s1} must reach roughly α_{s1}'s design titer. Heme from the best secreting strain, used at 0.45%, adds $1.1 per kg of beef analogue at the factory gate.

$/kg protein, US	BLG	OVA	ovomucoid	β-casein	lactoferrin	albumin	α-lac
best published strain	$43.0	$227	$479	$808	$413	$425	$1,848
base case (30 g/L; lactoferrin 10 g/L)	$21.2	$21.3	$21.3	$22.0	$168	$215	$33.8
base case, design-report capital	$16.4	$16.5	$16.5	$17.0	$130	$185	$25.0
industry benchmark (42 g/L)	$15.8	$15.9	$16.0	$16.4	$106	$205	$28.2
physical-limit best case	$4.1	$4.2	$4.3	$4.5	$56.9	$172	$5.2
incumbent, 2025$ per kg protein	$8.7–$17.8	$9.7–$21.2	$9.7–$21.2	$7.9–$18.7	$250–$1,585	$40.0–$70.0	$25.0–$129

Plants: 10 kt/yr (lactoferrin 250 t/yr, albumin and α-lac 1 kt/yr); physical limit at 100 kt/yr (lactoferrin 1.6 kt/yr, α-lac about 19 kt/yr). Albumin's incumbent: bulk bovine albumin (technical grade).

Where evidence would help most, in our judgment:

1. **Secretion without growth.** How much protein does a strain secrete per g of cells per hour when it barely grows, over days, at 1–10 m³? This rate sets both today's cost and the best case, for every protein here.
2. **Open fed-batch data.** Titer, productivity and sugar yield, measured **together**. For casein: any secreted titer above a few g/L, and whether the unphosphorylated protein makes cheese like dairy casein. For bovine α-lac and κ-casein: any secreted titer above 1 g/L.
3. **Real plant costs.** The capital, staff and running costs of recent PF plants would settle the vessel price and the fixed-cost rule. Aseptic vessels would add $9.8/kg, and the two accounting conventions differ by $4.7/kg on BLG.
4. **The design ceiling.** Workable starting volumes, feed concentrations and O_2 transfer at scale.

Animal welfare. The cost model does not say which protein matters most for animals, but the products differ. Whey is a by-product of cheese, so replacing it mostly lowers the whey price. Casein is what people buy milk for. Egg white (OVA, ovomucoid) carries far more animal-years per kg of protein than either.

EU. At Dutch prices for sugar, energy, wages and construction, the BLG plant costs about $3.1/kg more. Capital at 4% over 20 years, instead of 7.5% over 10, would cut $3.9/kg. These are cost effects on one plant, not policy recommendations.

Caveats. Costs exclude R&D, licensing, regulation, sales, start-up, failed batches and the premium of a first plant. Strain parameters rest on published designs and company-reported averages. The casein and lactoferrin titers come from other proteins or species. Only AI reviewers have audited the work since version 2. Full report, audit log and data: the `pf-tea` repository.